

WRITTEN TEST - INSTRUCTIONS

Candidate Name:

Assessor Name:

Qualification:

Python Programming

Units of Competency

SICIP-IT-PYP-01-G: Lead Small Team
SICIP-IT-PYP-02-S: Comply to Ethical Standards in the ICT Workplace
SICIP-IT-PYP-01-O: Perform Basic Python Programming
SICIP-IT-PYP-02-O: Perform Advanced Python Programming
SICIP-IT-PYP-03-O: Develop Web Applications Using Django
SICIP-IT-PYP-04-O: Perform Data Analysis Using Python

Assessment Centre:

Date of Assessment:

Time Of Assessment:

Instruction's:

Read and understand the direct Wins carefully:

- this written examination is based on the performance criteria from all the units of competency in **Python Programming**
- this assessment activity will be used to measure your underpinning knowledge
- write your answers on the paper provided
- answer all the questions as best as possible
- you have 1 hour to complete this test

Multiple Choice Question

1. **What is the output of the following code?**

```
x = 10
def modify():
    x = 20
    print(x)
modify()
print(x)
```

A. 10 and 10 B. 20 and 20 C. 20 and 10 D. Error

2. **What is the output?**

```
x = 5
y = 2
print(x ** y // 3)
```

A. 8 B. 8.33 C. 6 D. 25

3. **What is the output of the following slicing operation?**

```
text = "Programming"
print(text[-1:-7:-2])
```

A. gnm B. gim C. gmig D. nim

4. **What is the output?**

```
try:
    result = 10 / 0
except ZeroDivisionError:
    print("A")
else:
    print("B")
finally:
    print("C")
```

A. A only B. A followed by C C. B followed by C D. C only

5.	What is the output? <pre>numbers = [1, 2, 3] new_numbers = numbers new_numbers.append(4) print(numbers)</pre>	A. [1, 2, 3]	B. [1, 2, 3, 4]	C. [4, 1, 2, 3]	D. Error
6.	What is the output? <pre>a = [1, 2, 3] b = a.copy() b[0] = 10 print(a, b)</pre>	A. [10, 2, 3] [10, 2, 3]	B. [1, 2, 3] [10, 2, 3]	C. [1, 2, 3] [1, 2, 3]	D. Error
7.	What is the output of this list comprehension? <pre>result = [x ** 2 for x in range(6) if x % 2 != 0] print(result)</pre>	A. [1, 4, 9, 16, 25]	B. [0, 4, 16]	C. [1, 9, 25]	D. [1, 3, 5]
8.	What is the output? <pre>values = {1, 2, 2, 3, 3, 3} print(len(values))</pre>	A. 6	B. 5	C. 3	D. 2
9.	Which operation raises an error?	A. {1, 2} {2, 3}	B. {1, 2} & {2, 3}	C. {1, 2} - {2, 3}	D. {1, 2} + {2, 3}
10.	What is the output? <pre>numbers = (10, 20, 30, 40) a, *b, c = numbers print(a, b, c)</pre>	A. 10 [20, 30] 40	B. 10 (20, 30) 40	C. [10] [20, 30] [40]	D. Error
11.	Which statement about a frozenset is correct?	A. It permits duplicate values	B. Elements can be added using add()	C. It is an immutable set	D. It supports indexing
12.	What is the output? <pre>for i in range(1, 6): if i == 3: continue if i == 5: break print(i, end=" ")</pre>	A. 1 2 3 4	B. 1 2 4	C. 1 2 4 5	D. 1 2
13.	What is the output? <pre>def calculate(*args): return sum(args) / len(args) print(calculate(2, 4, 6, 8))</pre>	A. 4	B. 5	C. 5.0	D. 20
14.	What is the output? <pre>def update(value, items=[]): items.append(value) return items print(update(1)) print(update(2))</pre>	A. [1] [2]	B. [1] [1, 2]	C. [1, 2] [1, 2]	D. It raises an error.
15.	What is required before applying binary search to a collection?	A. The collection must contain integers only	B. The collection must be sorted	C. The collection must be a set	D. The collection must have an even number of elements

16.	What are the time and auxiliary space complexities of the following function? <pre>def process(values): result = [] for value in values: result.append(value * 2) return result</pre> A. Time $O(1)O(1)$, space $O(1)O(1)$ B. Time $O(n)O(n)$, space $O(1)O(1)$ C. Time $O(n)O(n)$, space $O(n)O(n)$ D. Time $O(n^2)O(n^2)$, space $O(n)O(n)$
17.	Which file is traditionally used to identify a directory as a regular Python package? A. package.py B. main.py C. <code>__init__.py</code> D. setup.txt
18.	Which Python module is most directly associated with low-level network communication? A. socket B. tkinter C. shutil D. pathlib
19.	What is the output of the following file-handling code? <pre>with open("sample.txt", "w") as file: file.write("Python") with open("sample.txt", "a") as file: file.write(" Programming") with open("sample.txt", "r") as file: print(file.read())</pre> A. Programming B. Python C. Python Programming D. An error occurs because the file is closed
20.	What is the output? <pre>class Parent: def show(self): return "Parent" class Child(Parent): def show(self): return super().show() + " Child" obj = Child() print(obj.show())</pre> A. Parent B. Child C. Parent Child D. An error occurs
Write the short Answer of the following questions (21-30)	
21.	Who developed Python, and when was it first released?
22.	What is the difference between a compiler and an interpreter?

23.	What are the rules for naming variables in Python?
24.	What is the difference between break, continue and pass?
25.	What is the difference between a list, tuple, set and dictionary?
26.	What is the difference between a stack and a queue?
27.	What is the difference between a Python module and a package?

28.	What is the difference between the w and a file mode?		
29.	What is a constructor in Python?		
30.	What are inheritance and method overriding?		
Feedback to Candidate:			
Assessment decisions for this assessment activity:			
☐ Competent		☐ Not Yet Competent	
Candidate's Signature:		Date:	
Assessor Signature:		Date:	